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RESEARCH ARTICLE

Generalization Analysis of Object Detection Under Controlled Synthetic Fog Severity Shifts in Satellite Imagery

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ABSTRACT Satellite object detection can be affected by visibility degradation caused by fog, particularly when target objects occupy only a small number of pixels. This study analyzes the generalization behavior of object detectors under controlled synthetic fog severity shifts in satellite imagery. Fog-synthesized images were generated using physically based fog synthesis (PBFS)-inspired synthetic fog generation with fog severity levels ranging from $\beta = 0.2$ to $\beta = 1.6$, and an 8×8 cross-evaluation was conducted using YOLOv10s and Ultralytics YOLO26s detectors. The results show that mismatch between training and testing fog severity leads to performance degradation, with an asymmetric tendency: models trained under dense fog conditions showed larger degradation when evaluated under lower fog severity conditions than models trained under lower fog severity conditions evaluated under dense fog. A balanced mixed-severity training setting was further evaluated under the same training budget to examine whether exposure to multiple fog severities reduces sensitivity to severity shifts. The results show smoother performance changes across fog severity levels compared with single-severity training, suggesting that mixed-severity exposure can reduce sensitivity to fog severity mismatch within the controlled synthetic fog setting.

INDEX TERMS Fog severity, synthetic fog, object detection, PBFS-inspired fog generation, satellite imagery, balanced mixed-severity training.

I. INTRODUCTION

Satellite object detection can be affected by visibility degradation under foggy conditions, particularly when target objects occupy only a small number of pixels. This issue is important in satellite imagery because small aircraft targets often have limited visual details, and reduced contrast or blurred boundaries can directly affect detection performance [3]. In operational imaging environments, visibility conditions may vary, making it necessary to understand how object detectors behave when training and testing conditions are not identical [5], [9].

Synthetic fog generation provides a controlled way to examine detector behavior under different visibility conditions. In this study, physically based fog synthesis (PBFS)-inspired synthetic fog generation is used to generate fog-synthesized satellite images with different fog severity levels [3]. This formulation allows the fog severity parameter β to be varied systematically, while keeping the underlying image content and bounding-box annotations unchanged. However, because the depth information is estimated from monocular satellite images, the generated fog images should be interpreted as controlled synthetic fog variations rather than exact reconstructions of real atmospheric fog [2], [3].

Previous studies on adverse-weather object detection have mainly focused on improving performance under fixed degradation conditions or on general robustness to environmental changes [5], [7], [8], [9]. In contrast, less attention has been given to the direction and degree of mismatch between training and testing fog severity [11], [12], [13], [14], [15]. A detector trained under a single fog severity condition may perform well under matched conditions but may not

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generalize consistently when evaluated under different fog severity levels. Therefore, a systematic cross-evaluation is needed to distinguish condition-specific performance from fog severity generalization.

To investigate this issue, this study conducts an 8×8 cross-evaluation across fog severity levels ranging from $\beta = 0.2$ to $\beta = 1.6$ using YOLOv10s and Ultralytics YOLO26s detectors [16].

Each single-severity model is evaluated under all fog severity conditions, allowing matched and mismatched training–testing combinations to be compared directly. This design makes it possible to analyze how performance changes as the gap between training and testing fog severity increases.

In addition, a balanced mixed-severity training setting is evaluated under the same training budget as the single-severity models. Rather than increasing the number of training samples, this setting incorporates multiple fog severity levels within a fixed data budget, enabling a fair comparison between condition-specific training and mixed-severity training. The objective of this study is not to claim dataset-independent generalization across all satellite imagery, but to provide a controlled analysis of fog severity mismatch and mixed-severity training within the considered synthetic fog setting.

II. METHODOLOGY

A. PBFS-INSPIRED SYNTHETIC FOG GENERATION

A PBFS-inspired synthetic fog generation process was applied to produce fog-synthesized images under controlled fog severity conditions [3]. The purpose of this process is not to reconstruct real atmospheric fog exactly, but to introduce systematic visibility degradation while preserving the image content and bounding-box annotations. Compared with simple image-level augmentation methods, this approach provides a controllable way to vary fog severity through the scattering coefficient β.

The fog generation process follows the atmospheric scattering formulation. The observed image is defined as

$$\boldsymbol{I}(\boldsymbol{x}) = \boldsymbol{J}(\boldsymbol{x})\boldsymbol{t}(\boldsymbol{x}) + \boldsymbol{A}(\mathbf{1} - \boldsymbol{t}(\boldsymbol{x})), \tag{1}$$

where J(x) denotes the scene radiance, A represents atmospheric light, and t(x) is the transmission map, which describes the portion of scene radiance that remains after attenuation. The transmission map is expressed as

$$\mathbf{t}(\mathbf{x}) = \boldsymbol{e}^{-\boldsymbol{\beta} \boldsymbol{d}(\boldsymbol{x})}, \tag{2}$$

where β denotes the fog severity parameter and d(x) represents the estimated relative depth. Larger β values reduce the transmission term and produce stronger visibility degradation. Depth information was estimated from each image using the MiDaS-based monocular depth estimation framework [2], which provides relative depth cues from a single RGB image. Although the estimated depth does not represent metric atmospheric distance, it provides spatial structure that can be used to apply non-uniform fog attenuation across the image. Therefore, objects and background regions are affected differently depending on their estimated relative depth values, resulting in spatially varying visibility degradation rather than uniform brightness reduction.

An example of the estimated depth map used in this process is shown in Fig. 1. The estimated depth maps were used to introduce spatially varying synthetic fog while preserving the overall scene structure, object locations, and bounding-box annotations of the satellite images. In this study, β ranges from 0.2 to 1.6, atmospheric light A is set to 0.98, and the depth scaling factor is fixed at 4.2. As β increases, the transmission term decreases exponentially, resulting in reduced contrast, blurred object boundaries, and partial loss of fine-grained visual features. These changes are relevant to satellite object detection because small aircraft targets often depend on local contrast, edge information, and boundary clarity for reliable detection.

Because monocular depth estimation on near-nadir satellite imagery does not directly represent true atmospheric path length, the generated fog images are used as controlled synthetic fog variations. Accordingly, this study focuses on relative detection performance trends under systematically controlled fog severity conditions.

B. DATASET CONSTRUCTION UNDER VARYING FOG SEVERITY

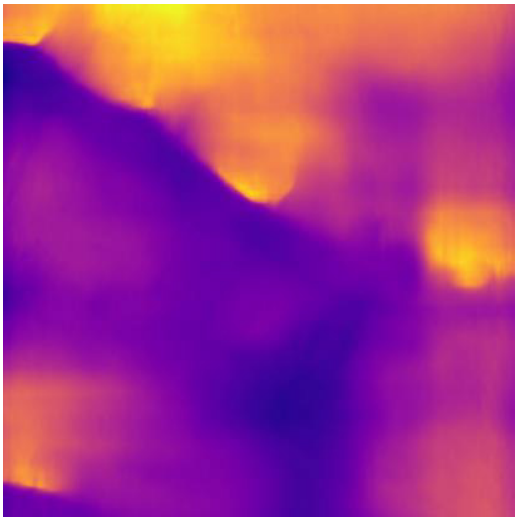
The experiments were conducted using the publicly available MAR20 dataset, a remote sensing military aircraft benchmark available through Roboflow Universe [10]. The dataset contains satellite images of military aircraft from multiple remote sensing scenes and includes 20 aircraft categories (A1–A20). Bounding-box annotations provided by Roboflow Universe were used without additional manual modification. During preprocessing, duplicated or highly similar scenes and images without valid aircraft objects were excluded to reduce redundancy and maintain consistent evaluation conditions. After preprocessing, 2,844 original images were used for training and 774 original images were used for validation. All images were resized to 640 × 640 during training. The dataset is available under the CC BY 4.0 license, and the detailed dataset information used in this study is summarized in Table 1.

The dataset was constructed using PBFS-inspired synthetic fog generation with the fog severity parameter β ranging from 0.2 to 1.6. For each fog severity condition, the training data consisted of paired original images and fog-synthesized images generated at the corresponding β level. Each dataset included 2,844 original images and 2,844 fog-synthesized images, resulting in 5,688 training samples per condition. This paired construction was used to expose each model to both clear and fog-synthesized samples while preserving the same object categories and annotation format across all training conditions.

For validation, the same 774 original validation images were used to generate fog-synthesized validation images at



(a)



(b)

FIGURE 1. Example of monocular depth estimation for PBFS-inspired synthetic fog generation: (a) original satellite image and (b) estimated depth map generated by MiDaS.

TABLE 1. Dataset summary of the MAR20 dataset.

Category	Description
Dataset	MAR20
Source	Roboflow Universe
Dataset Type	Remote sensing military aircraft dataset
Aircraft Categories	20 classes (A1–A20)
Image Source	Satellite imagery
Preprocessing	Duplicate and invalid image filtering
Base Training Images	2,844
Image Resolution	640×640
Base Validation Images	774
Annotation Type	Bounding box
Fog Generation Method	PBFS-inspired
Fog Severity Range	$\beta = 0.2 - 1.6$
License	*CC BY 4.0*

each fog severity level. By preserving the same image identities across all validation conditions, the evaluation isolates the effect of fog severity from variations in scene composition, object category distribution, and bounding-box annotations. This configuration enables direct comparison across different fog severity levels and supports the 8 × 8 cross-evaluation protocol used in this study.

The detailed dataset composition for each fog severity condition is summarized in Table 2.

TABLE 2. Dataset configuration for cross-evaluation under different fog severity levels.

Fog Severity Level (β)	Training Data Composition	Validation Data Composition
0.2	Original 2,844 + Fog-synthesized (β=0.2, 2,844)	Original validation 774 → Fog-synthesized β=0.2
0.4	Original 2,844 + Fog-synthesized (β=0.4, 2,844)	Original validation 774 → Fog-synthesized β=0.4
0.6	Original 2,844 + Fog-synthesized (β=0.6, 2,844)	Original validation 774 → Fog-synthesized β=0.6
0.8	Original 2,844 + Fog-synthesized (β=0.8, 2,844)	Original validation 774 → Fog-synthesized β=0.8
1.0	Original 2,844 + Fog-synthesized (β=1.0, 2,844)	Original validation 774 → Fog-synthesized β=1.0
1.2	Original 2,844 + Fog-synthesized (β=1.2, 2,844)	Original validation 774 → Fog-synthesized β=1.2
1.4	Original 2,844 + Fog-synthesized (β=1.4, 2,844)	Original validation 774 → Fog-synthesized β=1.4
1.6	Original 2,844 + Fog-synthesized (β=1.6, 2,844)	Original validation 774 → Fog-synthesized β=1.6

C. CROSS-EVALUATION SETUP

To analyze generalization across fog severity conditions, each single-severity model was evaluated on validation datasets generated at all fog severity levels. This produced an 8 × 8 cross-evaluation matrix, where each row represents the training fog severity and each column represents the testing fog severity.

Matched train–test conditions indicate condition-specific performance, whereas mismatched conditions reflect generalization capability under unseen fog severities. This matrix-based evaluation allows performance degradation patterns to be systematically examined as the gap between training and testing fog severity increases.

D. BALANCED MIXED-SEVERITY TRAINING

To evaluate the effect of exposure to multiple fog severity levels, a balanced mixed-severity training setting was introduced as a matched-budget multi-condition baseline. Unlike single-severity training, which trains each model under a fixed fog

severity condition, this setting trains a single model using samples from multiple fog severity levels.

The training dataset was constructed using the same total data budget as the single-severity settings, consisting of 2,844 original images and 2,844 fog-synthesized images. The fog-synthesized images were sampled across $\beta = 0.2$–1.6, with 355 images from each level from $\beta = 0.2$ to $\beta = 1.4$ and 359 images from $\beta = 1.6$, resulting in 5,688 training images. This configuration enables a controlled comparison between condition-specific training and multi-condition exposure under identical data-budget constraints.

E. MULTI-SEED STABILITY EVALUATION

To assess whether the balanced mixed-severity training results were dependent on a single training run, an additional multi-seed stability evaluation was conducted. For both YOLOv10s and YOLO26s, the balanced mixed-severity models were trained using three random seeds: 42, 43, and 44. The same dataset construction, training configuration, and evaluation protocol were used for all seeds. The resulting performance values were reported as mean ± standard deviation across the three seeds. This evaluation was used to examine run-to-run variability rather than to introduce a separate training strategy.

F. THREE-SPLIT VALIDATION CONSISTENCY ANALYSIS

To examine whether the reported validation results were dependent on a specific validation partition, a three-split validation consistency analysis was conducted. For each fog severity level, the validation set was divided into three non-overlapping subsets, denoted as split A, split B, and split C, with 258 images per subset. The same image identities were preserved across all fog severity levels to avoid subset-composition effects.

For each detector family, eight fog-severity-specific trained models were evaluated across eight fog-severity-specific validation conditions and three validation subsets, resulting in 384 validation runs in total. The detailed evaluation protocol is summarized in Table 3.

TABLE 3. Evaluation protocol for three-split validation consistency analysis.

Item	Setting
Detector families	YOLOv10s, YOLO26s
Training fog severities	β = 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6
Testing fog severities	β = 0.2, 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6
Validation subsets	Split A, Split B, Split C
Images per subset	258
Images per β	774
Total evaluations	2 × 8 × 8 × 3 = 384

III. EXPERIMENTAL SETUP

A. MODEL CONFIGURATION

In this study, YOLOv10s and Ultralytics YOLO26s one-stage object detection models were adopted to analyze detection performance under varying synthetic fog severity conditions. YOLOv10s was used as the primary detector due to its lightweight structure and efficient real-time detection capability. YOLO26s denotes the small variant of the Ultralytics YOLO26 model family, following a YOLO11-based architecture with native NMS-free end-to-end inference and removal of Distribution Focal Loss (DFL) [16]. Both models were trained and evaluated using identical dataset configurations and protocols to compare single-severity and balanced mixed-severity training settings while isolating the effect of fog severity changes.

B. TRAINING SETTINGS

All experiments were conducted under a consistent training configuration to ensure fair comparison across fog severity conditions. The models were trained in Google Colab using a single NVIDIA H100 GPU with 80 GB of memory. Training was performed for 200 epochs with a batch size of 32, an input image size of 640 × 640, and the SGD optimizer with an initial learning rate of 0.01.

For the multi-seed stability evaluation, the balanced mixed-severity models of YOLOv10s and YOLO26s were trained using three random seeds: 42, 43, and 44. The same dataset construction, training configuration, and evaluation protocol were applied across all seeds. The detailed training configuration is summarized in Table 4.

TABLE 4. Training environment and hyperparameters.

Category	Value
GPU (Hardware)	NVIDIA H100 (80 GB)
Platform	Google Colab
Epochs	200
Batch Size	32
Image Size	640 × 640
Optimizer	SGD
Learning Rate	0.01
Device	Single GPU
Random Seeds	42, 43, and 44
Logging	Local training logs and plots

C. EVALUATION METRICS

The object detection models were evaluated using precision, recall, mAP@0.5, and mAP@0.5:0.95. Precision and recall measure the correctness of predicted detections and the proportion of detected ground-truth objects, respectively. mAP@0.5 evaluates detection performance at an IoU threshold of 0.5, while mAP@0.5:0.95 averages performance across multiple IoU thresholds and was used as the primary metric in this study. Precision, recall, and mAP@0.5 were used as supplementary indicators.

To evaluate performance under varying fog severity conditions, the same evaluation metrics were consistently applied across all matched and mismatched train–test fog severity pairs. This enabled direct comparison of detection performance under controlled fog-severity shifts while maintaining an identical evaluation protocol across experiments.

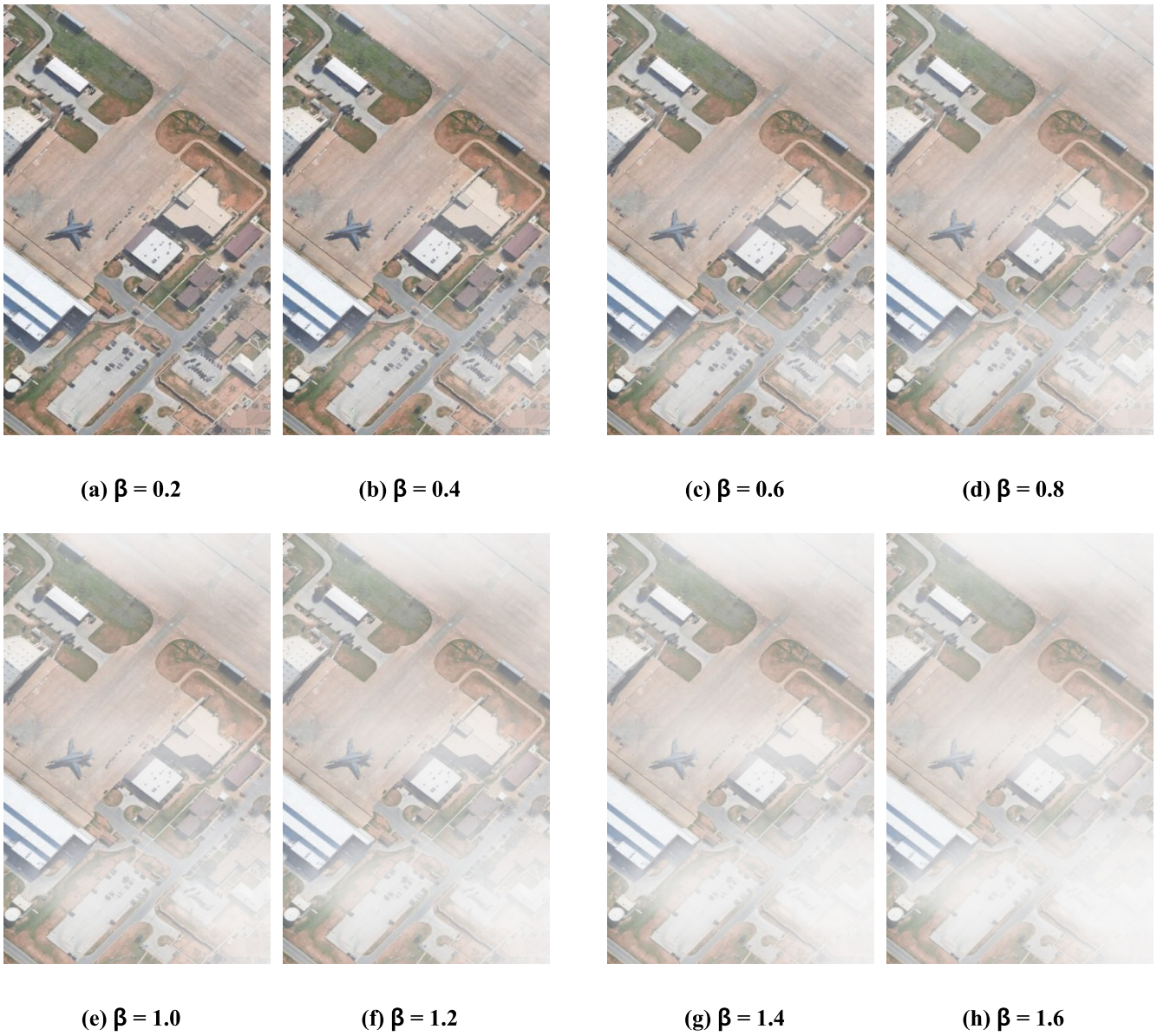


FIGURE 2. Example fog-synthesized satellite images under varying fog severity conditions ($\beta = 0.2$–1.6), where (a)–(h) correspond to increasing fog severity levels.

Multi-seed results are reported as mean ± standard deviation across three random seeds.

For the frequency-band analysis, average detection count, average top confidence, and average mean confidence were additionally used to assess detection stability under frequency manipulation.

The DCP-based dehazing comparison was evaluated using the same detection metrics, without additional retraining, to isolate the effect of preprocessing from model adaptation.

D. FOG IMAGE EXAMPLES

To illustrate the effect of fog severity on image appearance, fog-synthesized images were generated from the original satellite images using the PBFS-inspired synthetic fog generation process described in Section II-A.

The fog severity parameter β was varied from 0.2 to 1.6, as shown in Fig. 2. As β increases, visibility decreases, object boundaries become less distinguishable, and fine-grained visual details are partially lost. This visual progression confirms that the generated samples provide gradually increasing visibility degradation across the selected fog severity levels.

These examples show controlled synthetic fog severity variations used for the detection experiments, rather than exact reconstructions of real fog conditions.

IV. RESULTS AND ANALYSIS

A. CROSS-EVALUATION RESULTS

The cross-evaluation was conducted using eight models trained under different fog severity conditions ($\beta = 0.2$–1.6), with each model evaluated across all fog severity conditions

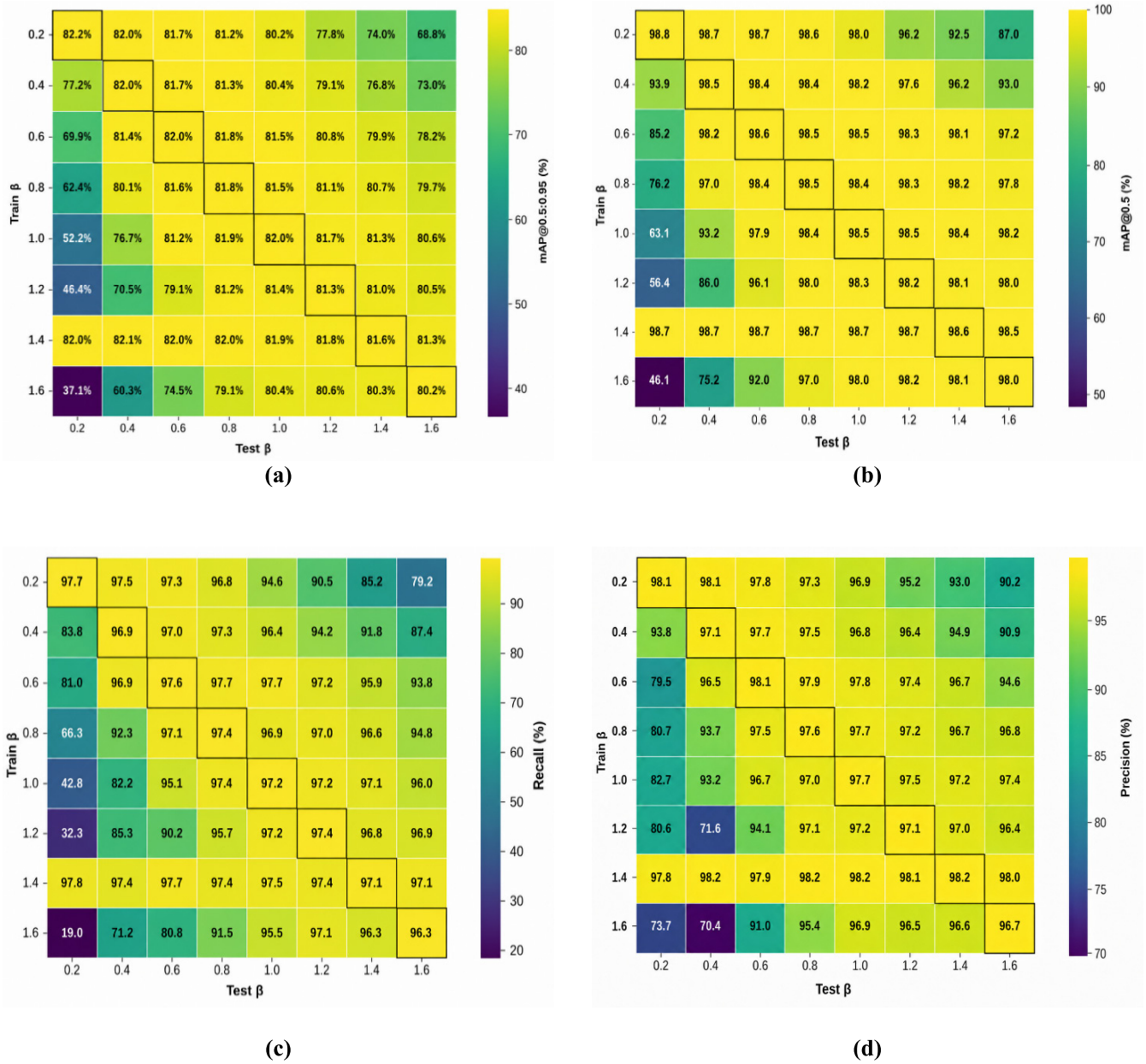


FIGURE 3. Cross-evaluation results of YOLOv10s under varying fog severity conditions: (a) mAP@0.5:0.95, (b) mAP@0.5, (c) recall, and (d) precision.

to analyze performance under both matched and mismatched settings. As shown in Fig. 3, YOLOv10s achieved strong performance under matched or near-matched conditions, while performance degradation was observed under fog severity mismatch conditions, particularly when models trained under dense fog were evaluated on low-severity fog images.

For example, a reduction in detection performance is observed when dense-fog-trained models are evaluated under low-severity fog conditions, whereas models trained under low-severity fog conditions maintain comparatively better generalization performance under dense fog conditions. The recall heatmaps indicate that performance reduction is primarily associated with missed detections, while precision remains relatively stable except under severe mismatch cases.

The YOLO26s results shown in Fig. 4 exhibit the same direction-dependent degradation pattern, but with a larger performance drop under the severe dense-to-low fog mismatch condition. The YOLO26s model trained at $\beta = 1.6$ achieved 5.4% mAP@0.5:0.95 and 7.7% recall at $\beta = 0.2$, whereas the $\beta = 0.2$-trained model maintained better detection performance under dense fog conditions. Since the 8×8 matrices in Figs. 3 and 4 were obtained from single-seed runs, the extreme mismatch cases are interpreted as representative observations rather than seed-averaged estimates.

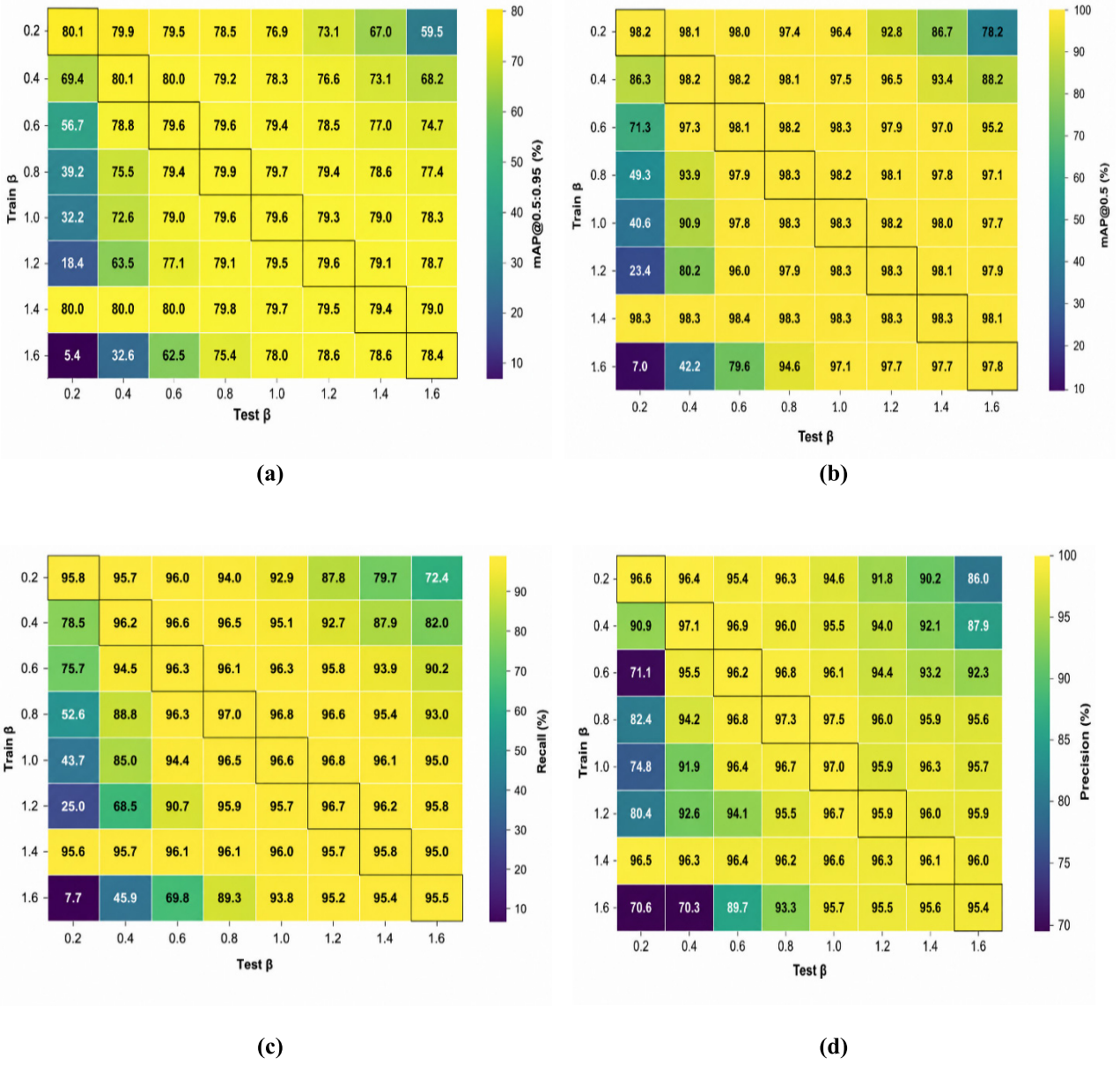


FIGURE 4. Cross-evaluation results of YOLO26s under varying fog severity conditions: (a) mAP@0.5:0.95, (b) mAP@0.5, (c) recall, and (d) precision.

B. FREQUENCY-BAND ANALYSIS

The cross-evaluation results indicate that performance degradation is associated with limited generalization across varying fog severity conditions. Models trained under a single fog severity level perform well under matched conditions but show reduced adaptability under mismatched conditions.

In particular, an asymmetric degradation pattern is observed. Models trained under higher fog severity conditions show larger performance reductions when evaluated under lower fog severity conditions, whereas models trained under lower fog severity conditions retain comparatively better generalization performance under dense fog scenarios. A similar tendency is also observed in YOLO26s, where dense-fog-trained models show limited generalization to low-severity fog conditions under extreme mismatch cases.

Although clear original images were included in every single-severity training set, the dense-fog-trained model produced near-zero detections under low-severity fog inputs. This suggests that the severe dense-to-low mismatch may involve confidence-calibration or operating-point instability under distribution shift, rather than only the loss of fine-grained visual cues.

This motivates examining frequency responses under matched and mismatched fog conditions. To further investigate this asymmetric behavior, frequency-band analysis was conducted using representative fog severity conditions with YOLOv10s.

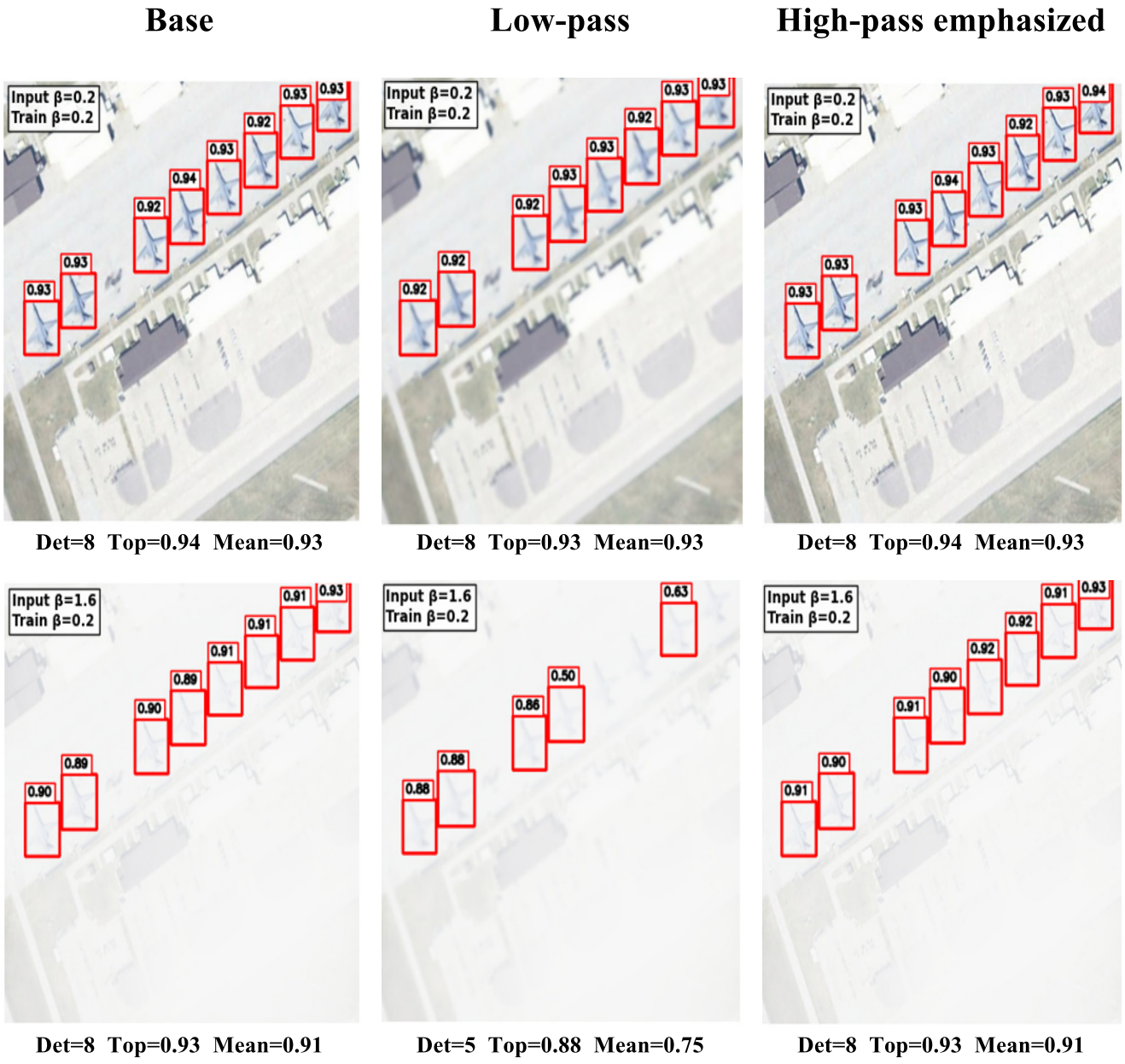


FIGURE 5. Frequency-band analysis under representative matched and mismatched fog-severity conditions using the YOLOv10s model.

In Fig. 5, detection stability is visualized under Base, Low-pass, and High-pass emphasized filtering conditions. Under the matched condition (train $\beta = 0.2$, input $\beta = 0.2$), the model maintains stable detection outputs across all frequency manipulations. The number of detected objects remains unchanged across the three filtering conditions, and the top and mean confidence scores show only minor variations. This indicates that the detector response is relatively stable when the training and input fog severities are consistent.

In contrast, under the mismatched dense-fog input condition (train $\beta = 0.2$, input $\beta = 1.6$), the detector response becomes more sensitive to frequency manipulation. In the Base and High-pass emphasized cases, the model still detects most aircraft objects with relatively high confidence. However, under Low-pass filtering, the number of detections decreases and the mean confidence score is noticeably reduced. This suggests that suppressing high-frequency and local structural cues, such as object boundaries and fine details, can further reduce detection stability under fog severity mismatch.

To examine whether this behavior is consistent across the entire evaluation set, the corresponding quantitative results are summarized in Table 5.

As shown in Table 5, the matched condition of train $\beta = 0.2$ and test $\beta = 0.2$ maintains stable average detection counts and confidence scores across all frequency conditions.

TABLE 5. Frequency-band analysis results under representative fog severity conditions.

Model	Train β	Test β	Filter	Images	Avg. detections	Avg. top confidence	Avg. mean confidence
YOLOv10s	*0.2*	*0.2*	*Base*	774	*5.75 ± 4.09*	*0.937 ± 0.062*	*0.923 ± 0.065*
			Low-pass		*5.68 ± 4.04*	*0.926 ± 0.088*	*0.905 ± 0.092*
			High-pass emphasized		*5.75 ± 4.10*	*0.938 ± 0.061*	*0.923 ± 0.065*
		1.6	*Base*		*4.62 ± 3.77*	*0.844 ± 0.248*	*0.813 ± 0.243*
			Low-pass		*3.70 ± 3.52*	*0.732 ± 0.343*	*0.699 ± 0.331*
			High-pass emphasized		*4.69 ± 3.79*	*0.855 ± 0.235*	*0.824 ± 0.230*
	1.6	*0.2*	*Base*		*0.13 ± 0.59*	*0.049 ± 0.192*	*0.047 ± 0.185*
			Low-pass		*0.12 ± 0.64*	*0.041 ± 0.179*	*0.038 ± 0.168*
			High-pass emphasized		*0.09 ± 0.51*	*0.031 ± 0.156*	*0.030 ± 0.149*
		1.6	*Base*		*5.70 ± 4.07*	*0.910 ± 0.070*	*0.884 ± 0.077*
			Low-pass		*5.57 ± 4.06*	*0.891 ± 0.136*	*0.858 ± 0.139*
			High-pass emphasized		*5.69 ± 4.08*	*0.907 ± 0.086*	*0.880 ± 0.091*

The average number of detections remains approximately 5.7, and the mean confidence remains around 0.91–0.92. A similar stable pattern is also observed under the matched dense-fog condition of train $\beta = 1.6$ and test $\beta = 1.6$, where the average number of detections remains approximately 5.6–5.7 with mean confidence values around 0.86–0.88.

In contrast, under the mismatched dense-fog testing condition (train $\beta = 0.2$, test $\beta = 1.6$), the average number of detections decreases from 4.62 in the Base condition to 3.70 under Low-pass filtering, while the average mean confidence decreases from 0.813 to 0.699. The High-pass emphasized condition partially preserves detection behavior, with 4.69 average detections and 0.824 average mean confidence. These results indicate that local edge-related cues remain important for detection under dense-fog inputs.

The severe reverse mismatch condition (train $\beta = 1.6$, test $\beta = 0.2$) shows a much larger degradation. Across Base, Low-pass, and High-pass emphasized inputs, the average number of detections remains very low, ranging from 0.09 to 0.13, and the average mean confidence remains below 0.05. This indicates that dense-fog-trained models have limited ability to generalize to low-severity fog inputs, regardless of frequency manipulation.

Overall, the frequency-band analysis supports the cross-evaluation results by showing that detection stability is maintained under matched fog severity conditions but decreases under mismatched conditions. The results further indicate that severe dense-to-clear mismatch results in substantially reduced detection responses, while dense-fog inputs evaluated by a low-severity-fog-trained model remain more sensitive to the loss of local structural cues.

C. BALANCED MIXED-SEVERITY TRAINING PERFORMANCE

To examine whether multi-condition exposure reduces sensitivity to training–testing mismatch, the balanced mixed-severity models described in Section II-D were evaluated across all fog severity levels. Unlike the single-severity models, which are optimized for one fixed fog condition, the balanced mixed-severity models were exposed to multiple fog severity levels during training. Therefore, this evaluation focuses on whether the models produce a more continuous performance trend when the testing fog severity changes.

The validation performance trends are illustrated in Fig. 6. Both YOLOv10s and YOLO26s show a gradual decrease in mAP@0.5:0.95 as fog severity increases, rather than the sharper degradation observed in the single-severity cross-evaluation results. This indicates that balanced mixed-severity training provides smoother and more stable performance across varying fog severity conditions, although performance degradation is still observed under severe fog conditions.

Notably, the performance curves do not show an abrupt collapse at high fog severity levels, suggesting that mixed-severity exposure improves stability across degraded visibility conditions. This result indicates a specialist–generalist tradeoff: single-severity models can achieve higher performance under matched fog conditions, whereas balanced mixed-severity training acts as a stability-oriented generalist baseline with more continuous performance under changing fog severity.

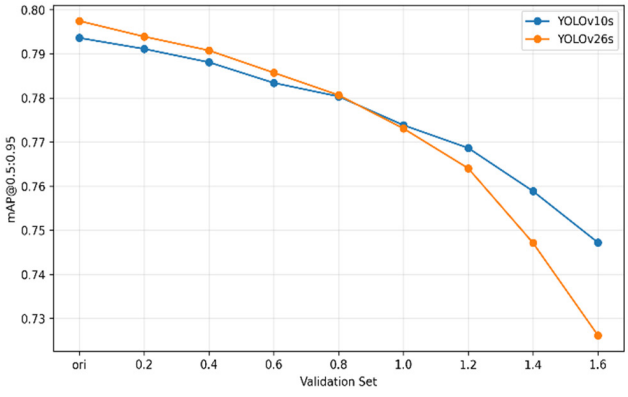


FIGURE 6. Validation mAP@0.5:0.95 of balanced mixed-severity YOLOv10s and YOLO26s models. The x-axis label "ori" denotes the original fog-free validation condition.

The corresponding quantitative results are summarized in Table 6.

For YOLOv10s, mAP@0.5:0.95 decreases from 79.4% under the original validation condition to 74.7% at $\beta = 1.6$, while recall decreases from 96.8% to 91.6%. Similarly, YOLO26s shows a decrease in mAP@0.5:0.95 from 79.7% to 72.6%, with recall decreasing from 96.1% to 87.9%.

Although performance degradation is still observed under severe fog conditions, the decline is more gradual compared to single-severity training.

TABLE 6. Validation performance of the balanced mixed-severity models under different fog severity conditions.

fog severity (β)	YOLOv10s mAP@0.5:0.95 (%)	YOLOv10s Recall (%)	YOLO26s mAP@0.5:0.95 (%)	YOLO26s Recall (%)
Original	79.4	96.8	79.7	96.1
0.2	*79.1*	*95.9*	*79.4*	*96.4*
0.4	*78.8*	*96.5*	79.1	95.3
0.6	*78.3*	*96.5*	78.6	95.2
0.8	*78.0*	*95.9*	78.1	94.5
1.0	*77.4*	*95.1*	77.3	93.9
1.2	*76.9*	*94.6*	76.4	92.1
1.4	*75.9*	*93.3*	74.7	91.0
1.6	*74.7*	*91.6*	72.6	87.9

D. MULTI-SEED STABILITY EVALUATION

To assess run-to-run stability, the balanced mixed-severity models were additionally trained using three random seeds: 42, 43, and 44. Table 7 reports the mean and standard deviation of mAP@0.5:0.95 and recall for YOLOv10s and YOLO26s. Both models show consistent decreasing trends as fog severity increases, indicating that the observed performance pattern was not caused by a single random initialization. For YOLOv10s, mAP@0.5:0.95 decreased from 79.2 ± 0.2% at $\beta = 0.2$ to 74.8 ± 0.4% at $\beta = 1.6$, while recall decreased from 96.6 ± 0.2% to 92.4 ± 1.3%. The standard deviation of mAP@0.5:0.95 remained within 0.4 percentage points across all fog severity levels, indicating stable performance under repeated training with different random seeds. For YOLO26s, mAP@0.5:0.95 decreased from 79.1 ± 0.3% at $\beta = 0.2$ to 72.6 ± 1.3% at $\beta = 1.6$, and recall decreased from 95.9 ± 0.1% to 88.7 ± 1.8%. Although YOLO26s followed the same overall decreasing trend, its variability became larger under severe fog conditions, particularly at $\beta = 1.4$ and $\beta = 1.6$.

Overall, the multi-seed evaluation supports the interpretation that the balanced mixed-severity performance trend reflects a reproducible tendency under the controlled synthetic fog setting, rather than an artifact of a particular seed. The consistent decreasing trend across both detector variants further indicates that the observed behavior is stable under repeated training with different random seeds.

TABLE 7. Multi-seed evaluation results.

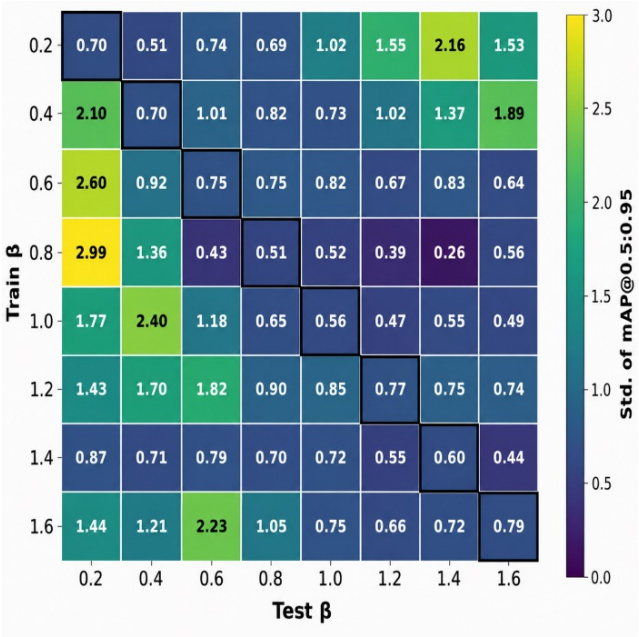
Fog Severity Level (β)	Model	mAP@0.5:0.95 (%)	Recall (%)
0.2	YOLOv10s	79.2 ± 0.2	96.6 ± 0.2
0.4		78.8 ± 0.2	96.1 ± 0.3
0.6		78.4 ± 0.1	96.0 ± 0.3
0.8		78.1 ± 0.2	95.8 ± 0.3
1.0		77.6 ± 0.1	95.3 ± 0.3
1.2		76.9 ± 0.1	94.8 ± 0.4
1.4		76.1 ± 0.2	94.0 ± 0.7
1.6		74.8 ± 0.4	92.4 ± 1.3
0.2	YOLO26s	79.1 ± 0.3	95.9 ± 0.1
0.4		78.7 ± 0.5	95.6 ± 0.3
0.6		78.2 ± 0.6	94.9 ± 0.3
0.8		77.7 ± 0.5	94.5 ± 0.5
1.0		77.1 ± 0.6	93.6 ± 1.2
1.2		76.1 ± 0.8	92.6 ± 1.7
1.4		74.6 ± 1.1	91.0 ± 1.3
1.6		72.6 ± 1.3	88.7 ± 1.8

E. THREE-SPLIT VALIDATION CONSISTENCY RESULTS

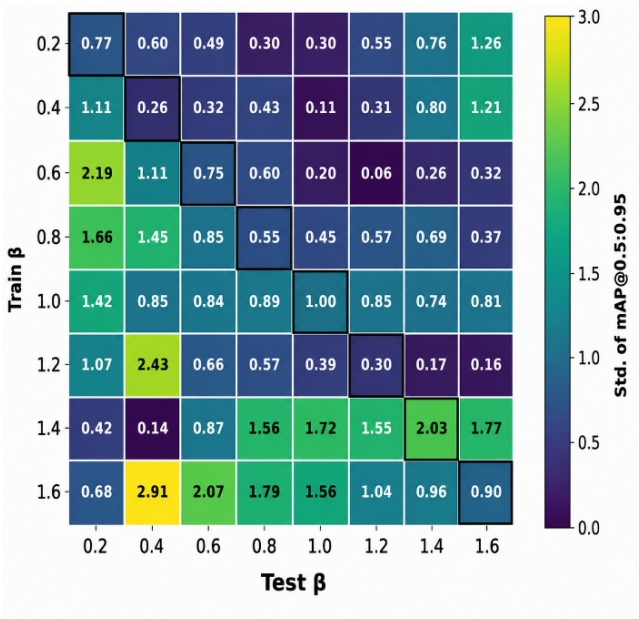
To evaluate whether the cross-evaluation results were dependent on a specific validation partition, a three-split validation consistency analysis was conducted. Following the evaluation protocol in Table 3, the eight models trained under specific fog severity levels were evaluated across the eight fog severity validation conditions over three validation subsets (split A, split B, and split C) for each detector family, resulting in 384 validation runs in total. Fig. 7 presents the standard deviation heatmaps of mAP@0.5:0.95 across split A, split B, and split C for YOLOv10s and YOLO26s. These heatmaps were systematically derived from the 384 individual validation outcomes by computing the standard deviation across the three validation subsets for each train–test pair. Overall, both detector families showed limited split-to-split variation for most train–test fog severity pairs.

For YOLOv10s, all diagonal cases showed low standard deviation values ranging from 0.51 to 0.79 percentage points. YOLO26s also showed low diagonal variation in most matched conditions, ranging from 0.26 to 1.00 percentage points, except for the $\beta = 1.4$ matched condition, where the standard deviation increased to 2.03 percentage points.

Low-variation regions were also observed in many near-diagonal and moderately mismatched conditions. In particular, YOLOv10s maintained relatively low variation in several regions with test $\beta \geq 0.6$, while YOLO26s showed stable variation in many conditions with test β from 0.6 to 1.2. Relatively higher standard deviation values appeared mainly in severe train–test mismatch regions. For YOLOv10s, the largest variations occurred at train $\beta = 0.8$ and test $\beta = 0.2$, with 2.99 percentage points, and at train $\beta = 0.6$ and test $\beta = 0.2$, with 2.60 percentage points. For YOLO26s, higher variations were observed at train $\beta = 1.6$ and test $\beta = 0.4$, with 2.91 percentage points, and at train $\beta = 1.2$ and test $\beta = 0.4$, with 2.43 percentage points.



(a)



(b)

FIGURE 7. Standard deviation heatmaps of mAP@0.5:0.95 for three-split validation consistency: (a) YOLOv10s and (b) YOLO26s.

These results suggest that the observed variation is primarily associated with severe fog severity mismatch rather than arbitrary subset selection. Therefore, the three-split validation analysis provides additional subset-level evidence that the reported performance tendencies are not artifacts of a single validation partition. Although this analysis does not replace external dataset validation, the generally low-variance patterns observed in both YOLOv10s and YOLO26s strengthen the internal reliability of the dataset-based evaluation protocol.

F. DCP-BASED DEHAZING COMPARISON

As a supplementary image-level preprocessing comparison, additional experiments used dark channel prior (DCP)-based dehazing on the $\beta = 1.6$ validation dataset to examine if dehazing recovers detection performance under severe synthetic fog. The $\beta = 1.6$ condition represents the most severe synthetic fog setting in this study. DCP served as a classical image-level preprocessing reference, not a satellite-specific adverse-weather detection baseline. Therefore, this comparison provides a limited reference for dehazing-based preprocessing, without establishing DCP as an optimized baseline for satellite object detection. No additional retraining or dataset-specific DCP parameter tuning was performed.

(a)



(b)

FIGURE 8. Example results of DCP-based dehazing applied to $\beta = 1.6$ validation images: (a) fog-synthesized image and (b) DCP-dehazed image.

Fig. 8 presents a representative example under the most severe synthetic fog condition, where (a) shows the fog-synthesized image and (b) shows the corresponding DCP-dehazed result. The quantitative evaluation was conducted on the $\beta = 1.6$ validation dataset and is summarized in Table 8. Although the dehazing process partially enhances the visibility of aircraft structures and surrounding background regions compared with the fog-synthesized image, such visual improvement does not necessarily correspond to improved detector responses.

The DCP-dehazed images were evaluated using the balanced mixed-severity YOLOv10s and YOLO26s models without additional retraining. The results in Table 8 show detector-dependent behavior.

TABLE 8. Detection performance after dcp-based dehazing under severe synthetic fog.

Method	Model	mAP@0.5:0.95 (%)	Recall (%)
Balanced Mixed-Severity	YOLOv10s	74.7	91.6
DCP + Balanced Mixed-Severity	YOLOv10s	73.9	90.4
Balanced Mixed-Severity	YOLO26s	72.6	87.9
DCP + Balanced Mixed-Severity	YOLO26s	73.1	88.3

For YOLOv10s, mAP@0.5:0.95 decreased from 74.7% to 73.9%, and recall decreased from 91.6% to 90.4% after DCP preprocessing. In contrast, YOLO26s showed a marginal improvement in mAP@0.5:0.95 from 72.6% to 73.1%, and recall from 87.9% to 88.3%. These mixed results indicate that classical image-level dehazing does not consistently recover detection performance under severe synthetic fog conditions. One possible reason is that DCP modifies image contrast and color distribution at the pixel level, while the detectors were not retrained or adapted to the dehazed image distribution. Therefore, visual enhancement alone may not provide a stable detection benefit, especially when the preprocessing output differs from the distribution observed during training.

V. CONCLUSION

This study investigated the generalization behavior of object detection models under controlled fog severity shifts in satellite imagery using PBFS-inspired synthetic fog generation. Through an 8×8 cross-evaluation across multiple fog severity levels, the results showed that models trained under a single fog condition achieved strong performance under matched conditions but exhibited reduced generalization capability under mismatched conditions. In particular, the degradation was affected by the direction of the fog severity shift, indicating that training–testing mismatch is an important factor in fog-related detection performance.

The balanced mixed-severity training results showed that exposure to multiple fog severity levels can reduce sensitivity to severity mismatch and provide smoother performance changes across fog severity levels compared with single-severity training. Similar trends observed in both YOLOv10s and YOLO26s further support that the observed tendency is consistent across the YOLO-based detector variants considered in this study.

The additional multi-seed and three-split validation analyses indicate that the reported trends were not artifacts of a single training run or a specific validation partition. In addition, the DCP-based dehazing comparison showed detector-dependent and inconsistent effects, suggesting that classical image-level dehazing alone does not reliably recover detection performance under severe synthetic fog without retraining or adaptation to the dehazed image distribution.

Overall, the results highlight the importance of considering fog severity variability during training when object detectors are expected to operate under changing visibility conditions. However, because the experiments were conducted using synthetic fog generated from a single satellite dataset, the findings should be interpreted as internally validated trends within the MAR20-based evaluation setting rather than dataset-independent evidence of real-world fog generalization. Future work should extend the analysis to real foggy remote sensing imagery, additional satellite datasets, and broader adverse-weather baselines to further evaluate the generality of the observed trends.

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